AWS Infrastructure Documentation

CampusChain: Scaling the Future of Higher Education

This official documentation details the Amazon Web Services architecture powering the CampusChain ecosystem. Each service has been selected for its industry-leading scalability, security, and integration capabilities.

Amazon EC2 (Elastic Compute Cloud)

1. Why we used it
To provide a scalable, high-performance hosting environment for the CampusChain web application and backend services.

2. Working
Virtual compute instances that provide resizable capacity in the cloud. We use pre-configured images (AMIs) to ensure consistent deployments across developer environments.

3. Real-World Example
Netflix uses EC2 to scale its streaming platform to handle peak traffic during major releases without service interruption.

Amazon S3 (Simple Storage Service)

1. Why we used it
For high-durability storage of immutable transaction receipts and digital artifacts generated on the blockchain.

2. Working
Object storage built to store and retrieve any amount of data from anywhere. Data is stored across multiple Availability Zones to ensure 99.999999999% durability.

3. Real-World Example
Dropbox leverages S3 to provide users with a secure and redundant cloud storage experience for billions of files.

Amazon SNS (Simple Notification Service)

1. Why we used it
To implement an instant alerting system that notifies administrators via email/SMS whenever the Fraud Engine detects high-risk transactions.

2. Working
A fully managed Pub/Sub messaging service. Topic-based distribution allows one alert to reach multiple administrative endpoints simultaneously.

3. Real-World Example
Uber uses SNS to send real-time push notifications, ride receipts, and authentication codes to millions of users globally.

Amazon CloudWatch

1. Why we used it
To monitor system health, application logs, and custom metrics like total token volume and API error rates in real-time.

2. Working
Collects monitoring and operational data in the form of logs and metrics, providing actionable insights into performance bottlenecks.

3. Real-World Example
Airbnb uses CloudWatch Alarms to automatically trigger recovery actions and scaling when latency thresholds are exceeded.

Amazon DynamoDB

1. Why we used it
To maintain a permanent, high-speed audit log of all security events and fraud detections with single-digit millisecond latency.

2. Working
A serverless NoSQL database that automatically scales to handle throughput requirements without any manual server management.

3. Real-World Example
Amazon.com uses DynamoDB to handle the massive high-velocity data surges during Prime Day sales events.